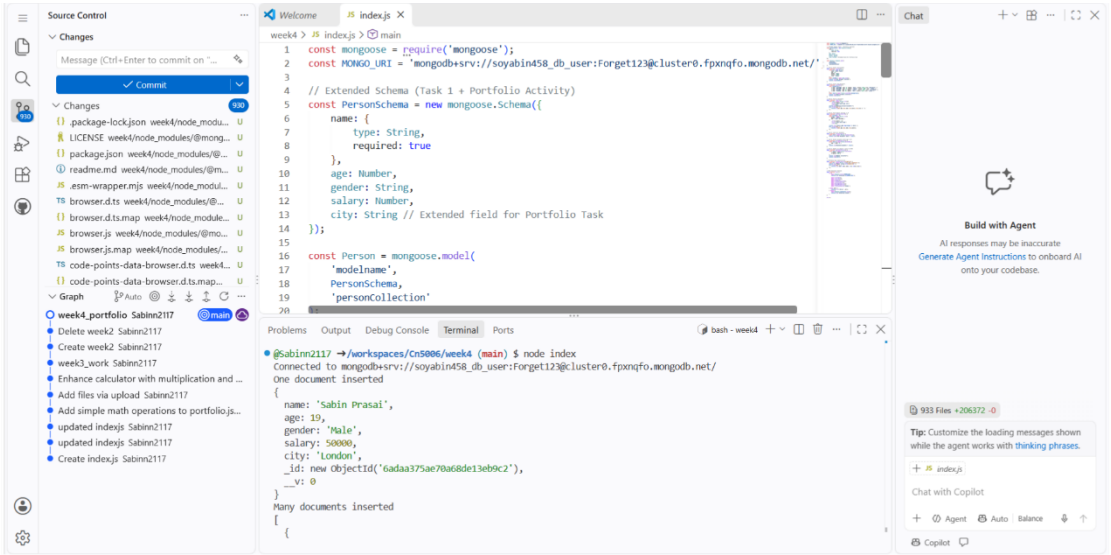
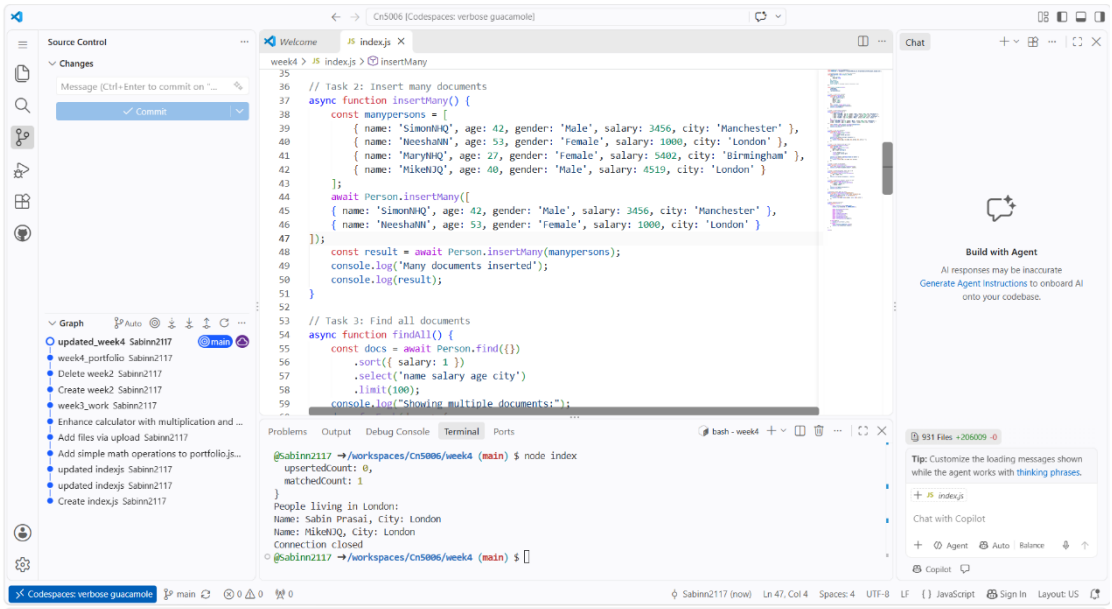
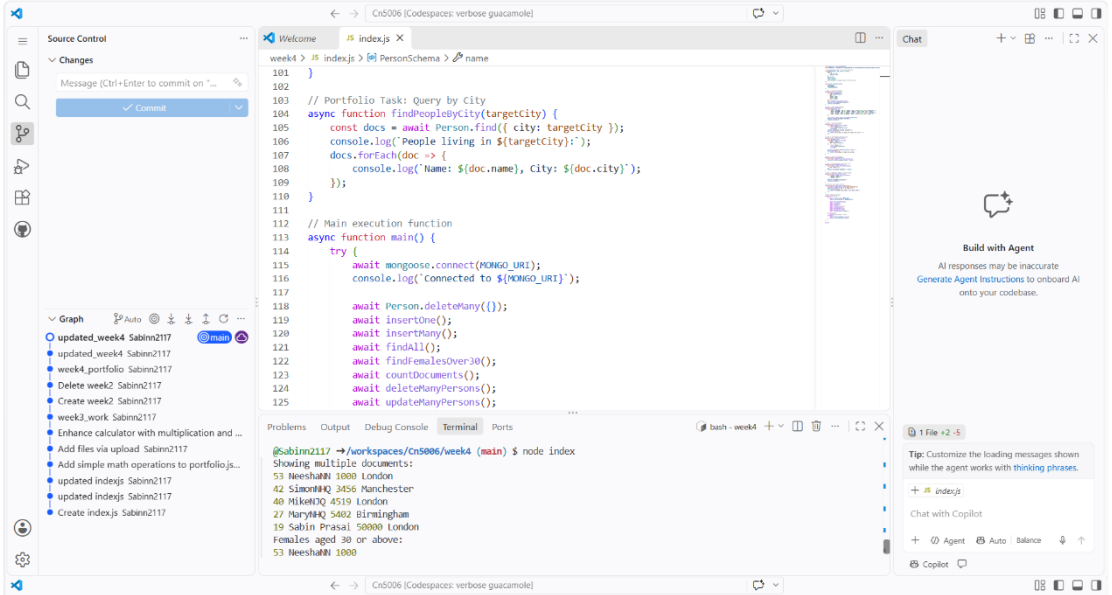
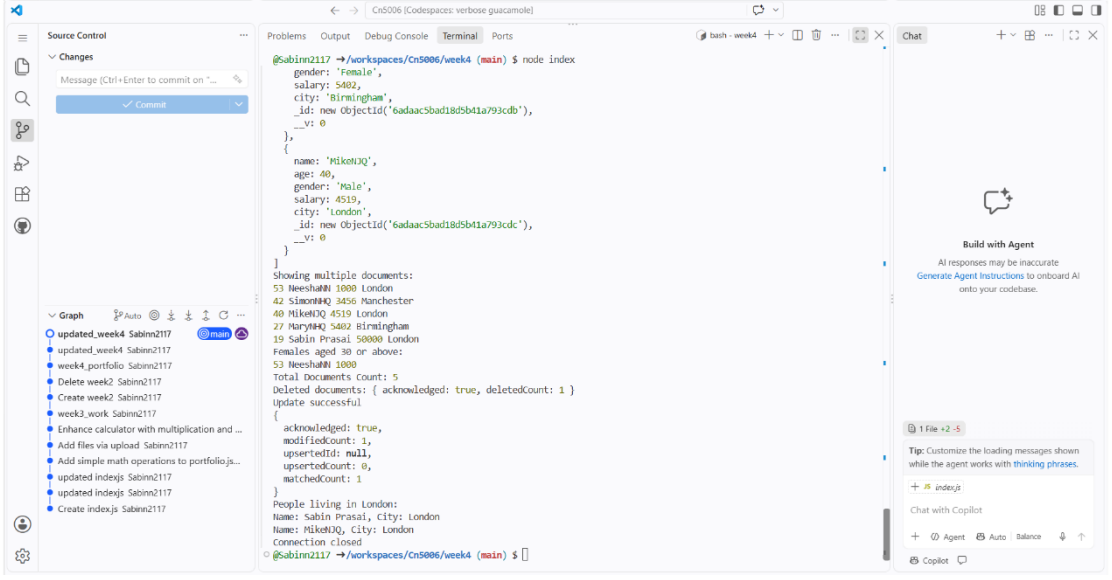
REFLECTION_WEEK4

These lab sessions helped me with hands-on experience in basic programming concepts in the JavaScript programming language and fundamentals of the Node.js runtime environment, from procedural programming to module-based software design. From the basic parameterized functions such as EmployeeInfo, I was able to process and format the dynamic data, all the way to using the modern ES6 arrow functions (() => {}) for writing concise functional code. I looked at making my own local CommonJS modules (Person and StudentInfo), and learning how to use exports and module.exports to contain reusable helper methods and classes that were loaded into index.js using require(). I also leveraged core modules of Node.js to get direct access to system information and diagnostics: Memory allocation, platform information, system uptime etc. Last, I combined these skills by creating a customized script for the PortfolioTask.js application that shows academic information and live execution time-stamps, runs string manipulation algorithms to generate a verification number by adding the last four digits of my student ID, and records the execution evidence and reflective problem-solving in a structured format.




Cn5006 [Codespaces: verbose guacamole]
Source Control
Changes
Message (Ctrl+Enter to commit on "...
Commit
Graph
updated_week4 Sabinn2117
week4_portfolio Sabinn2117
Delete week2 Sabinn2117
Create week2 Sabinn2117
week3_work Sabinn2117
Enhance calculator with multiplication and ...
Add files via upload Sabinn2117
Add simple math operations to portfolio.js...
updated indexjs Sabinn2117
updated indexjs Sabinn2117
Create index.js Sabinn2117
Welcome
index.js
week4 > index.js > insertMany
// Task 2: Insert many documents
async function insertMany() {
const manypersons = [
{ name: 'SimonNHQ', age: 42, gender: 'Male', salary: 3456, city: 'Manchester' },
{ name: 'NeeshaNN', age: 53, gender: 'Female', salary: 1000, city: 'London' },
{ name: 'MaryNHQ', age: 27, gender: 'Female', salary: 5402, city: 'Birmingham' },
{ name: 'MikeNJQ', age: 40, gender: 'Male', salary: 4519, city: 'London' }
];
await Person.insertMany([
{ name: 'SimonNHQ', age: 42, gender: 'Male', salary: 3456, city: 'Manchester' },
{ name: 'NeeshaNN', age: 53, gender: 'Female', salary: 1000, city: 'London' }
]);
const result = await Person.insertMany(manypersons);
console.log('Many documents inserted');
console.log(result);
}
// Task 3: Find all documents
async function findAll() {
const docs = await Person.find({})
.sort({ salary: 1 })
.select('name salary age city')
.limit(100);
console.log("Showing multiple documents:");
Problems Output Debug Console Terminal Ports
bash - week4
@Sabinn2117 →/workspaces/Cn5006/week4 (main) $ node index
upsertedCount: 0,
matchedCount: 1
}
People living in London:
Name: Sabin Prasai, City: London
Name: MikeNJQ, City: London
Connection closed
@Sabinn2117 →/workspaces/Cn5006/week4 (main) $
Chat
Build with Agent
AI responses may be inaccurate
Generate Agent Instructions to onboard AI onto your codebase.
931 Files +206009 -0
Tip: Customize the loading messages shown while the agent works with thinking phrases.
index.js
Chat with Copilot
Agent Auto Balance
Copilot
Codespaces: verbose guacamole
main
Sabinn2117 (now) Ln 47, Col 4 Spaces: 4 UTF-8 LF { } JavaScript Sign In Layout: US


Cn5006 [Codespaces: verbose guacamole]
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updated indexjs Sabinn2117
Create index.js Sabinn2117
Welcome
index.js
week4 > index.js > PersonSchema > name
}
// Portfolio Task: Query by City
async function findPeopleByCity(targetCity) {
const docs = await Person.find({ city: targetCity });
console.log(`People living in ${targetCity}:`);
docs.forEach(doc => {
console.log(`Name: ${doc.name}, City: ${doc.city}`);
});
}
// Main execution function
async function main() {
try {
await mongoose.connect(MONGO_URI);
console.log(`Connected to ${MONGO_URI}`);
await Person.deleteMany({});
await insertOne();
await insertMany();
await findAll();
await findFemalesOver30();
await countDocuments();
await deleteManyPersons();
await updateManyPersons();
Problems Output Debug Console Terminal Ports
bash - week4
@Sabinn2117 →/workspaces/Cn5006/week4 (main) $ node index
Showing multiple documents:
53 NeeshaNN 1000 London
42 SimonNHQ 3456 Manchester
40 MikeNJQ 4519 London
27 MaryNHQ 5402 Birmingham
19 Sabin Prasai 50000 London
Females aged 30 or above:
53 NeeshaNN 1000
Chat
Build with Agent
AI responses may be inaccurate
Generate Agent Instructions to onboard AI onto your codebase.
1 File +2 -5
Tip: Customize the loading messages shown while the agent works with thinking phrases.
index.js
Chat with Copilot
Agent Auto Balance
Copilot


Cn5006 [Codespaces: verbose guacamole]
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Enhance calculator with multiplication and ...
Add files via upload Sabinn2117
Add simple math operations to portfolio.js...
updated indexjs Sabinn2117
updated indexjs Sabinn2117
Create index.js Sabinn2117
Problems Output Debug Console Terminal Ports
bash - week4
@Sabinn2117 →/workspaces/Cn5006/week4 (main) $ node index
gender: 'Female',
salary: 5402,
city: 'Birmingham',
_id: new ObjectId('6adaac5bad18d5b41a793cdb'),
__v: 0
},
{
name: 'MikeNJQ',
age: 40,
gender: 'Male',
salary: 4519,
city: 'London',
_id: new ObjectId('6adaac5bad18d5b41a793cdc'),
__v: 0
}
]
Showing multiple documents:
53 NeeshaNN 1000 London
42 SimonNHQ 3456 Manchester
40 MikeNJQ 4519 London
27 MaryNHQ 5402 Birmingham
19 Sabin Prasai 50000 London
Females aged 30 or above:
53 NeeshaNN 1000
Total Documents Count: 5
Deleted documents: { acknowledged: true, deletedCount: 1 }
Update successful
{
acknowledged: true,
modifiedCount: 1,
upsertedId: null,
upsertedCount: 0,
matchedCount: 1
}
People living in London:
Name: Sabin Prasai, City: London
Name: MikeNJQ, City: London
Connection closed
@Sabinn2117 →/workspaces/Cn5006/week4 (main) $
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Agent Auto Balance
Copilot